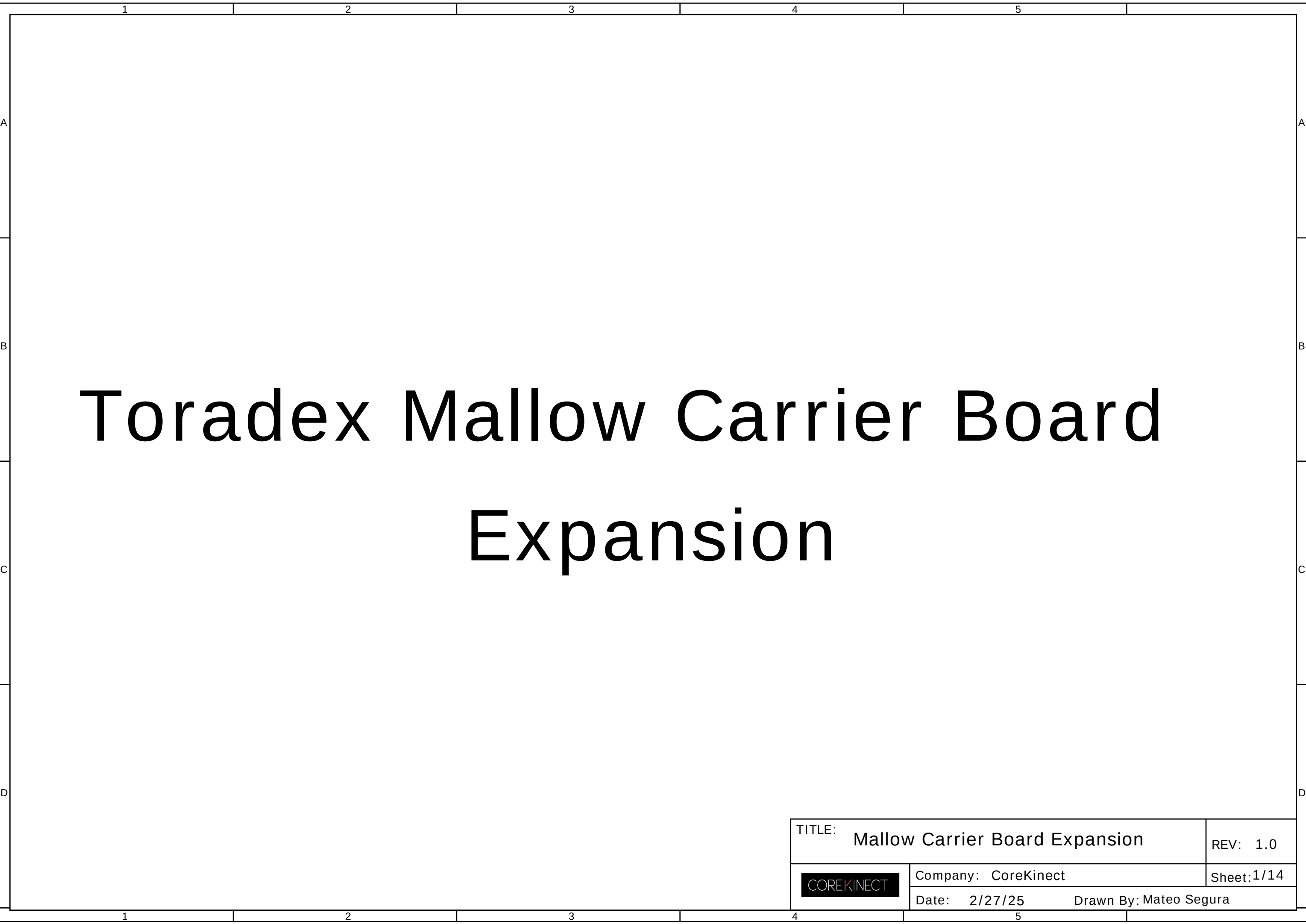
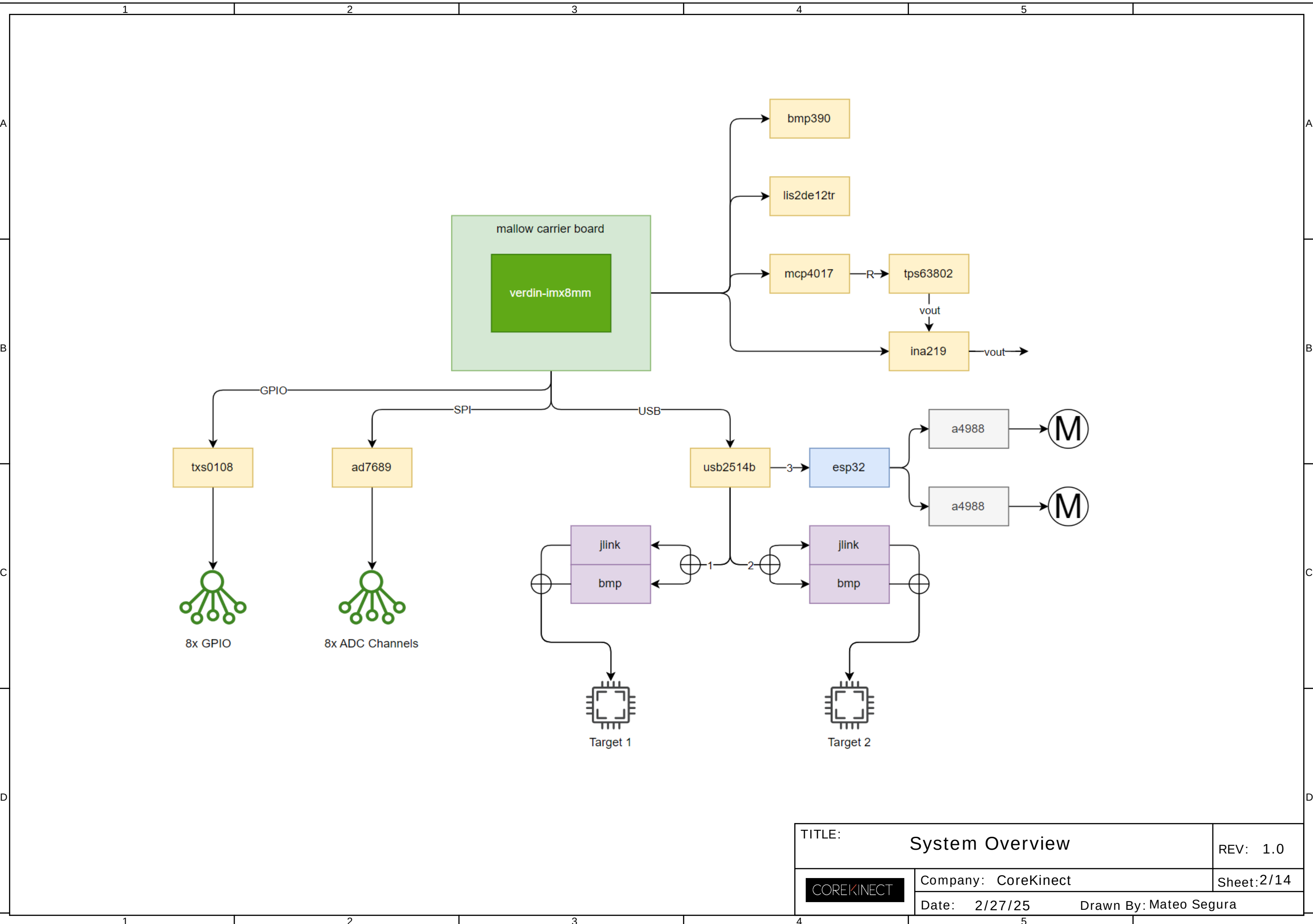




Toradex Mallow Carrier Board Expansion

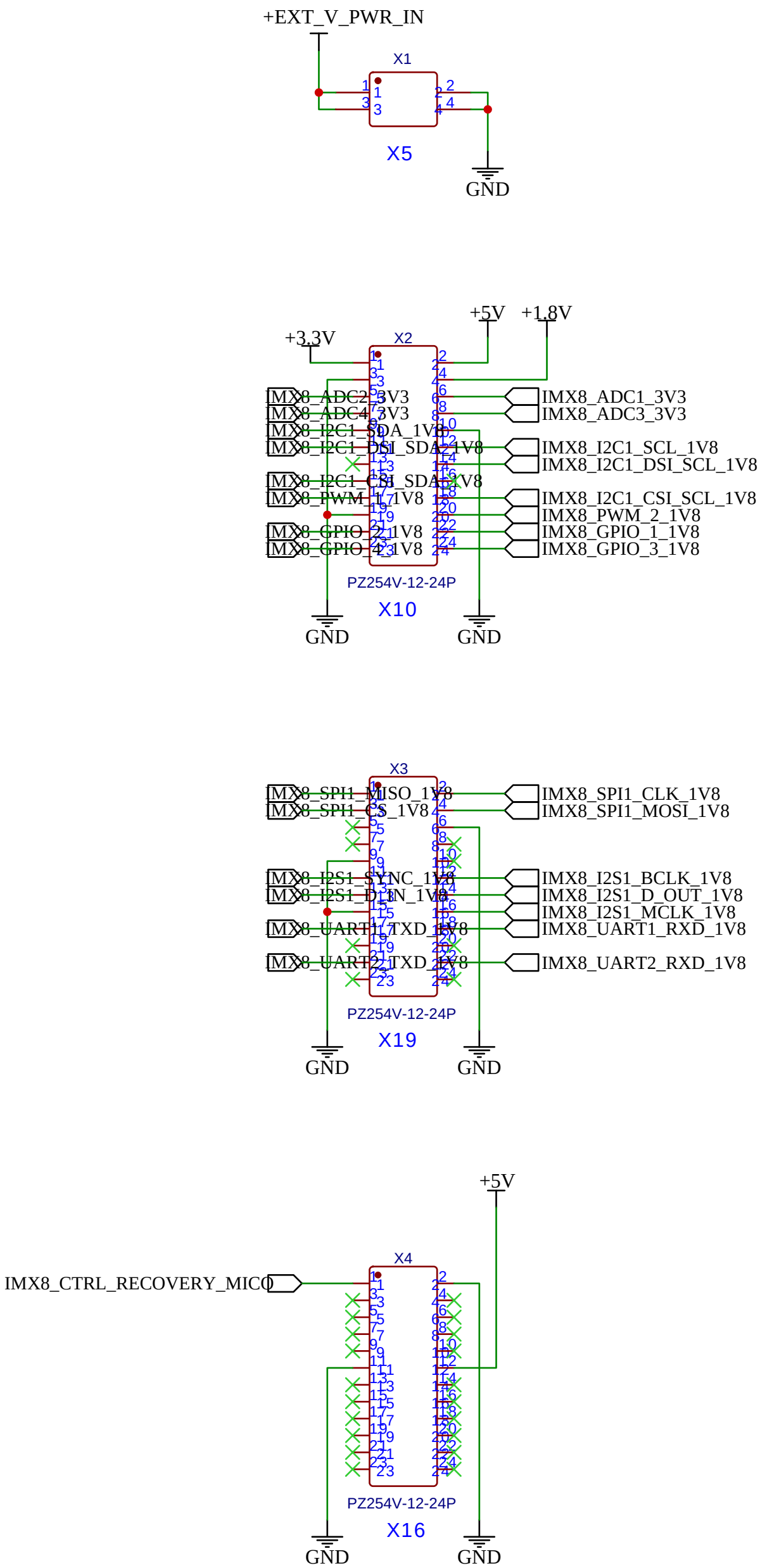
TITLE: Mallow Carrier Board Expansion		REV: 1.0
	Company: CoreKinect	Sheet:1/14
	Date: 2/27/25 Drawn By: Mateo Segura	



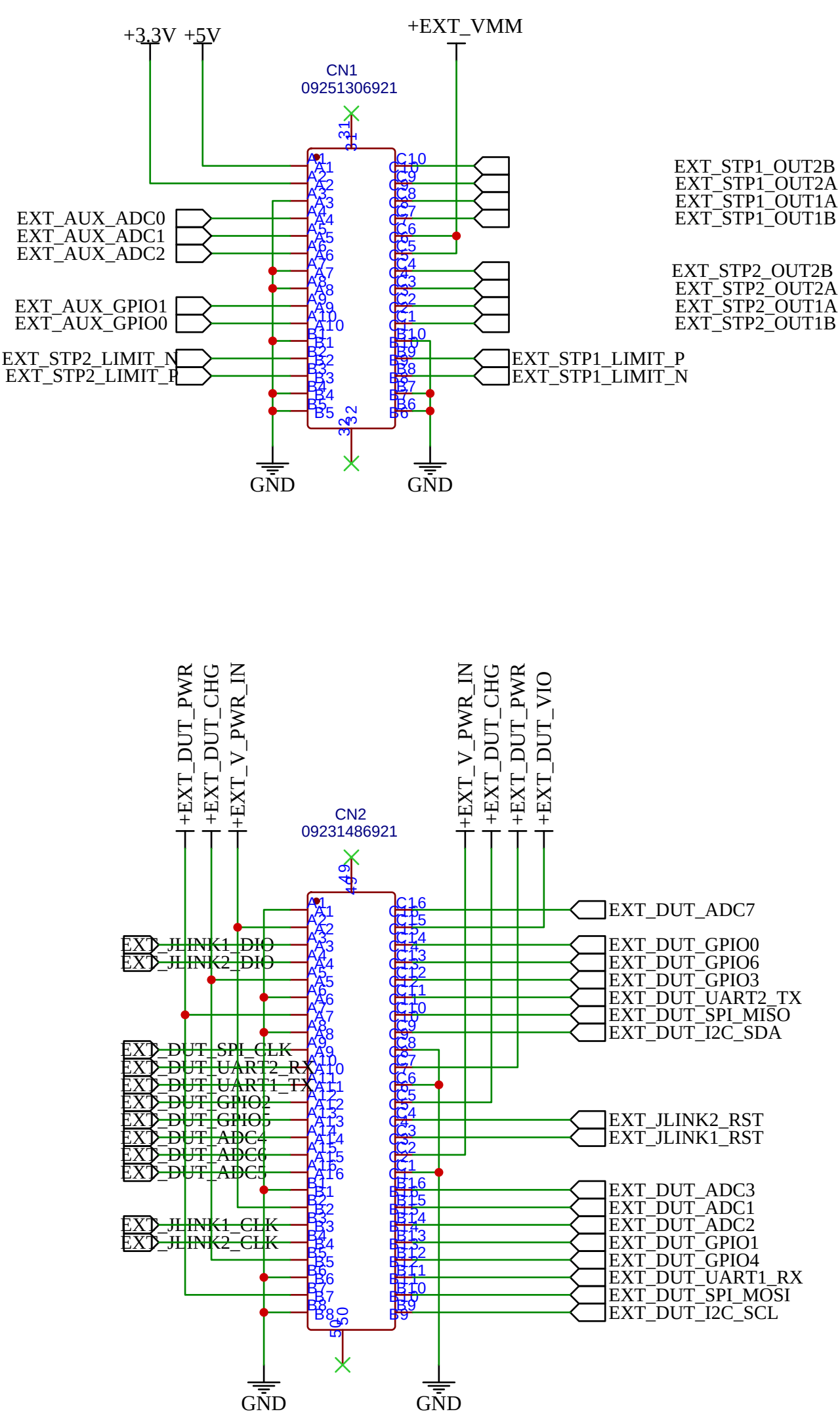
TITLE: System Overview		REV: 1.0
COREKINECT	Company: CoreKinect	Sheet:2/14
	Date: 2/27/25	Drawn By: Mateo Segura

Block diagram doesnt explain inputs/outputs

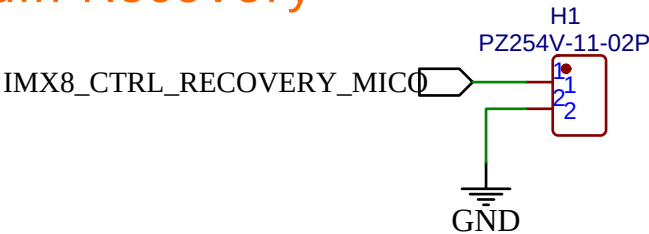
Mallow Carrier Board Connections



External Connectors

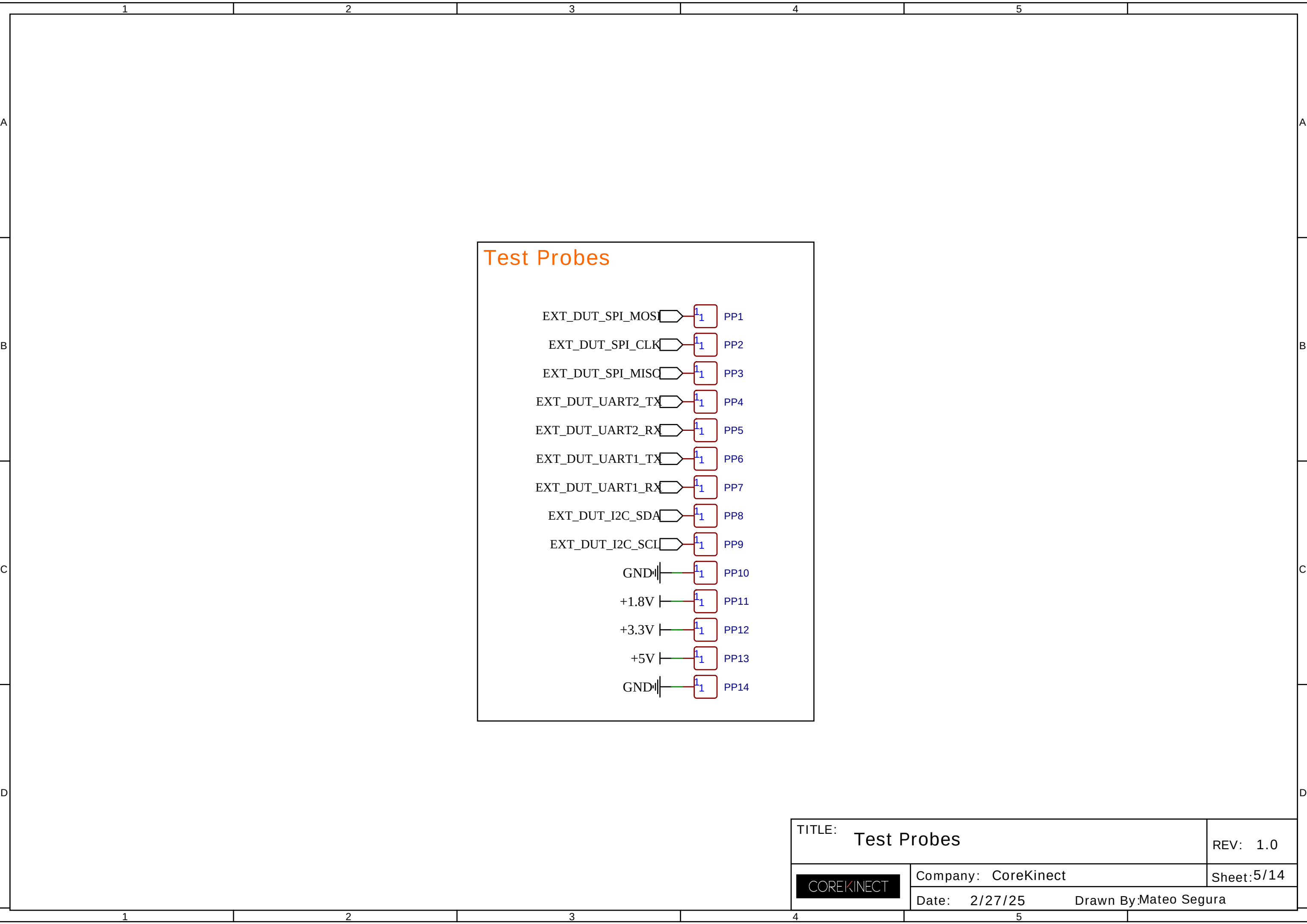


Verdin Recovery



Closing this header allows the SoM to be flashed with the TeZi image installer

TITLE: Connectors		REV: 1.0
COREKINECT	Company: CoreKinect	Sheet:3/14
	Date: 2/27/25	Drawn By: Mateo Segura



TITLE:Test Probes

REV:1.0

COREKINECT

Company:CoreKinect

Sheet:5/14

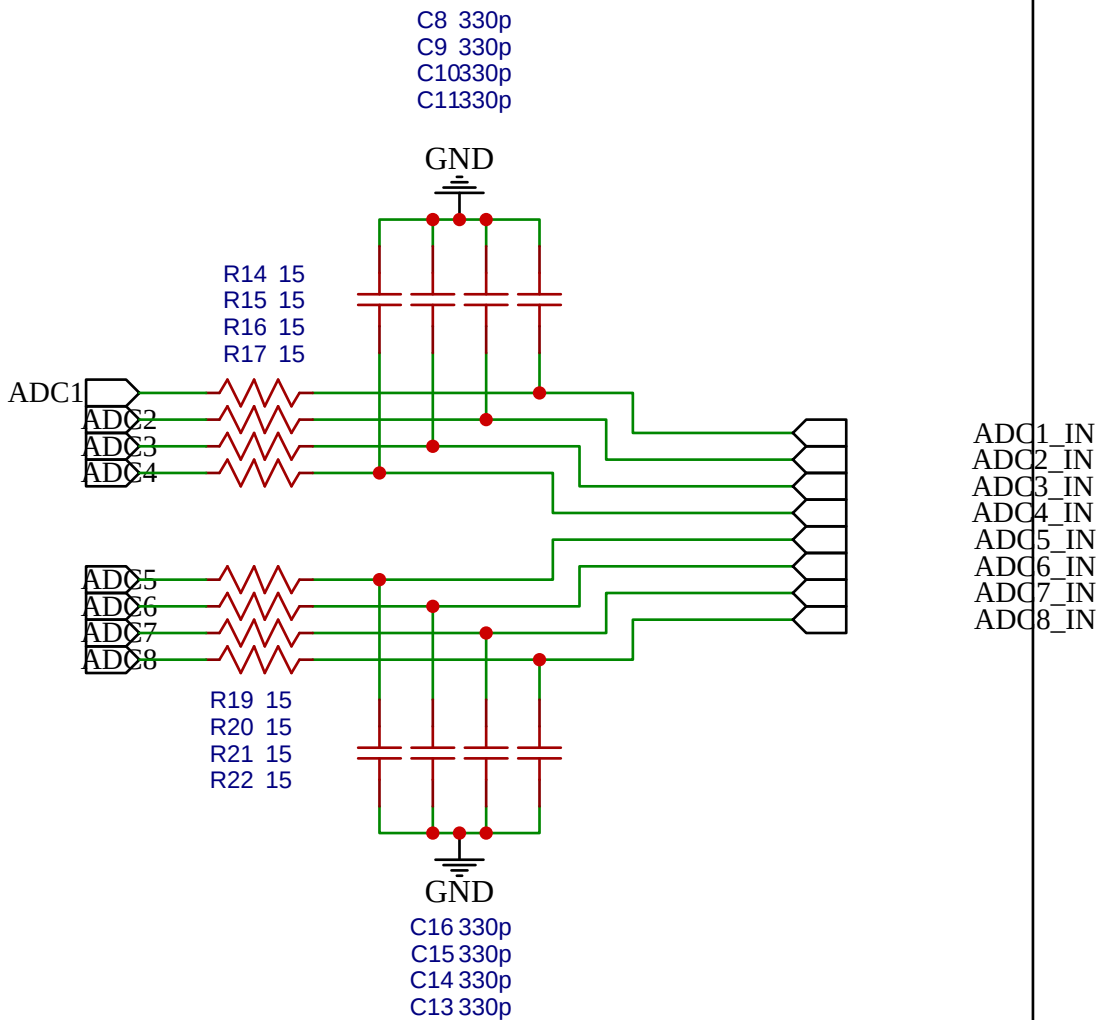
Date:2/27/25

Drawn By:Mateo Segura

12345

D

Final Stage Signal Conditioning



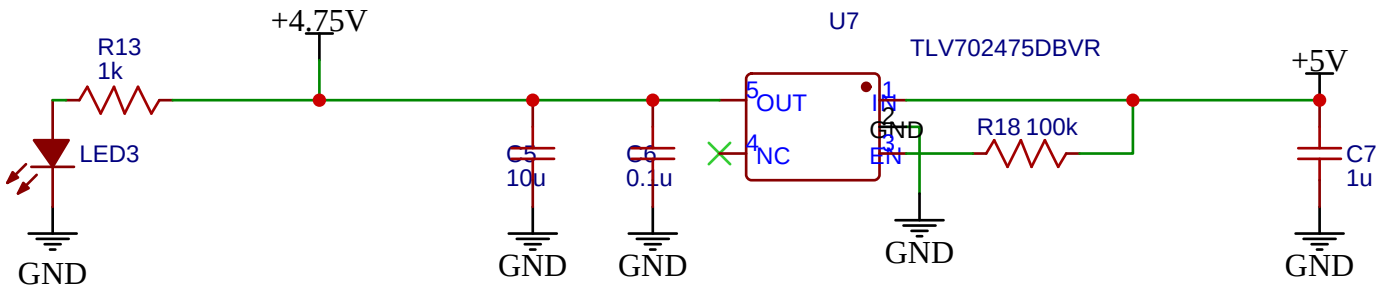
Internal Connections

ADC_I2C_SDA_3V3 BMX8_I2C1_SDA_3V3
ADC_I2C_SCL_3V3 BMX8_I2C1_SCL_3V3

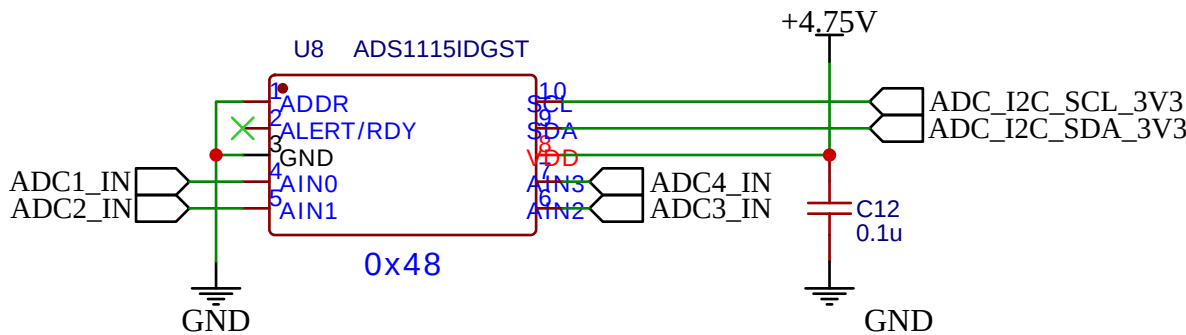
Test Points

TP12  +4.75V

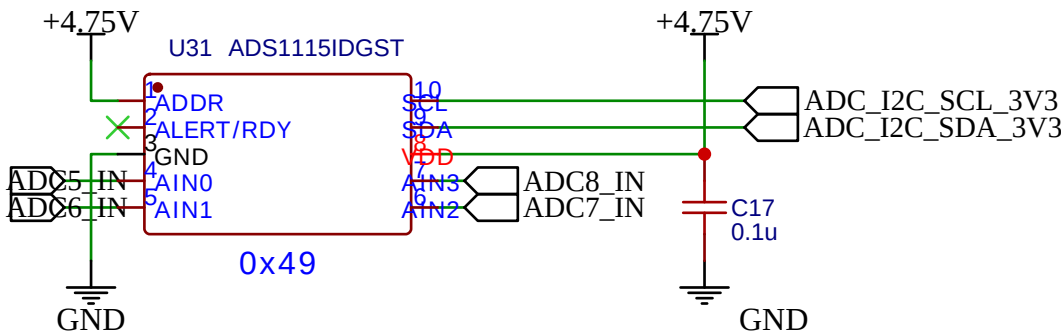
Clean Power Supply



Analog to Digital Converters



ADDR PIN CONNECTION	TARGET ADDRESS
GND	1001000b
VDD	1001001b
SDA	1001010b
SCL	1001011b



TITLE: Analog to Digital Converter

REV: 1.0

COREKINET

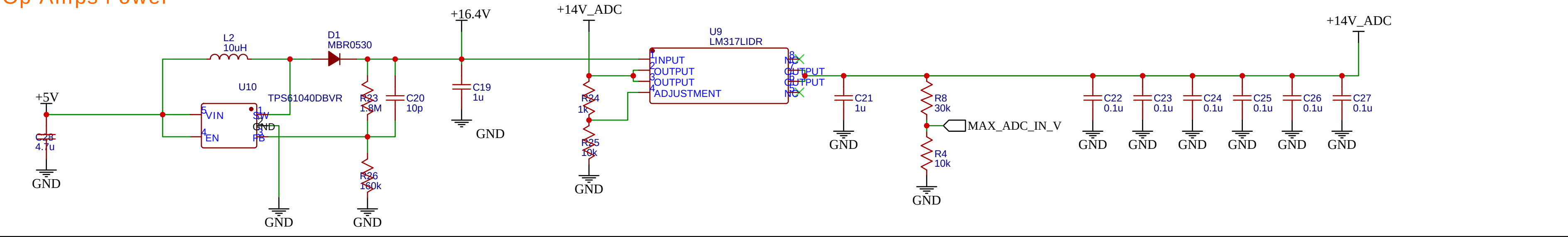
Company: CoreKinect

Sheet:6/14

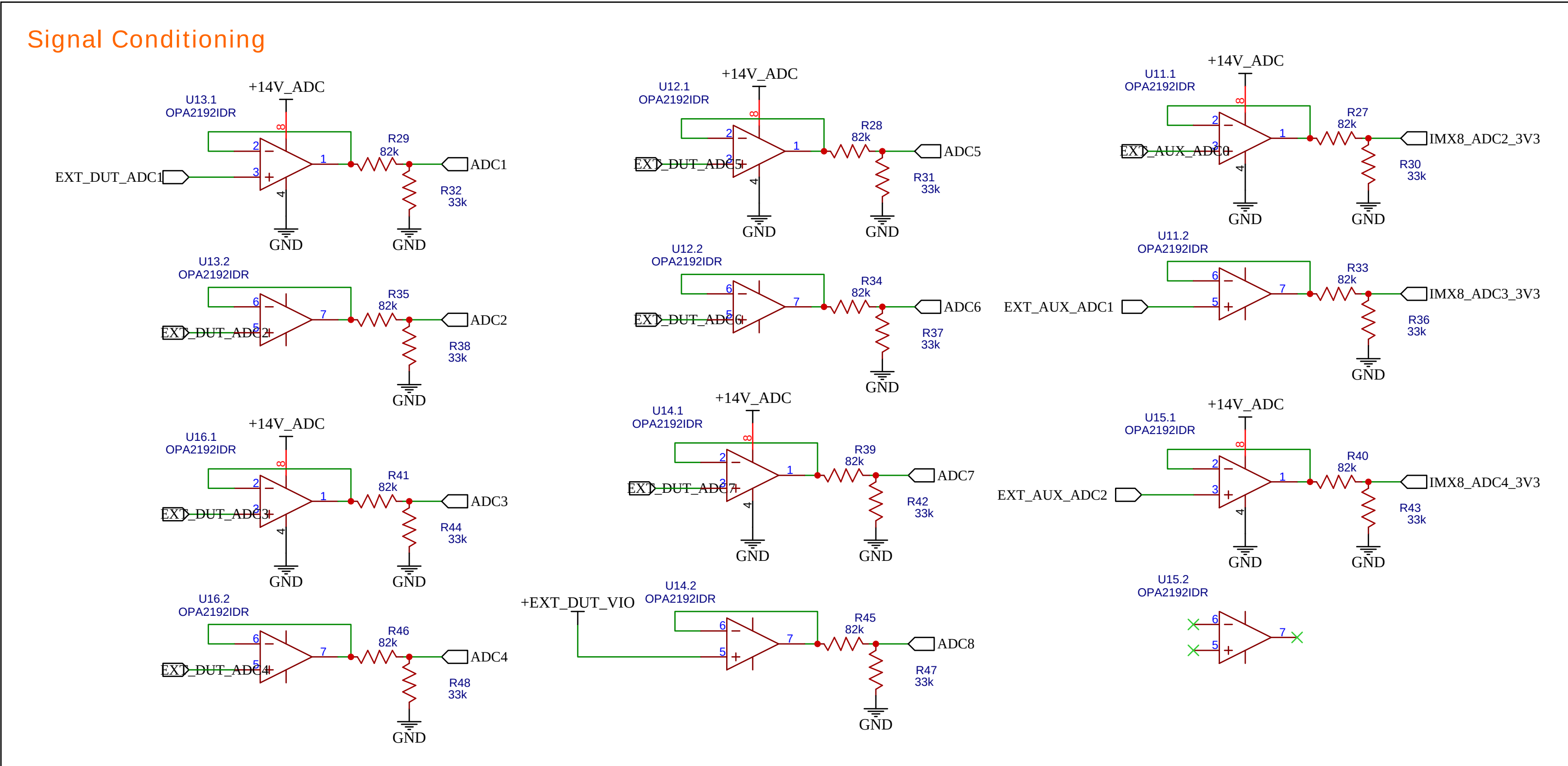
Date: 2/27/25

Drawn By: Mateo Segura

Op Amps Power



Signal Conditioning



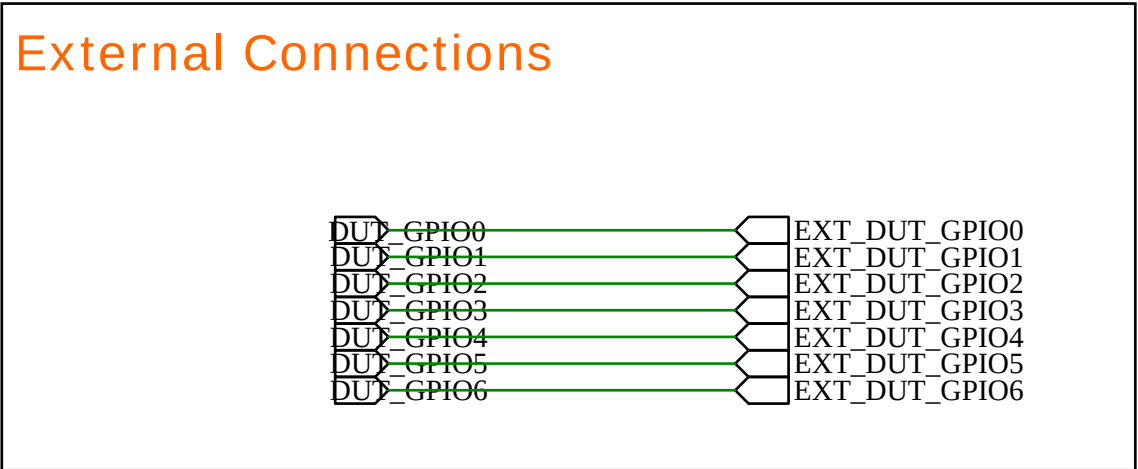
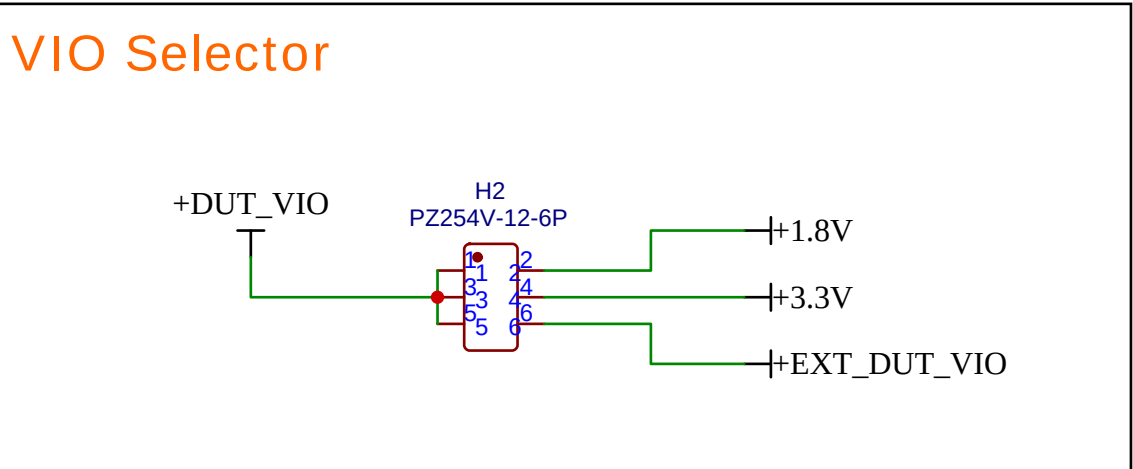
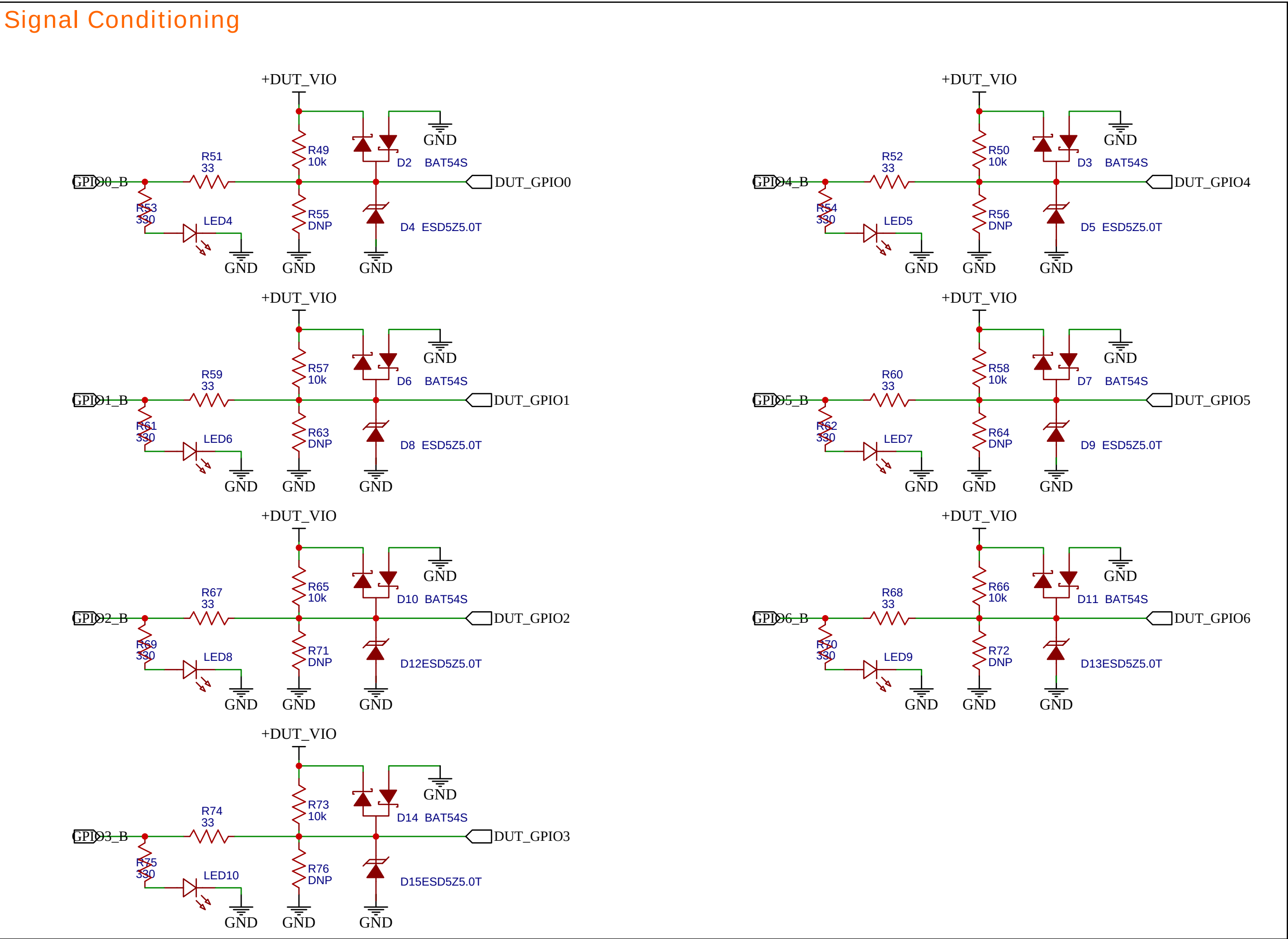
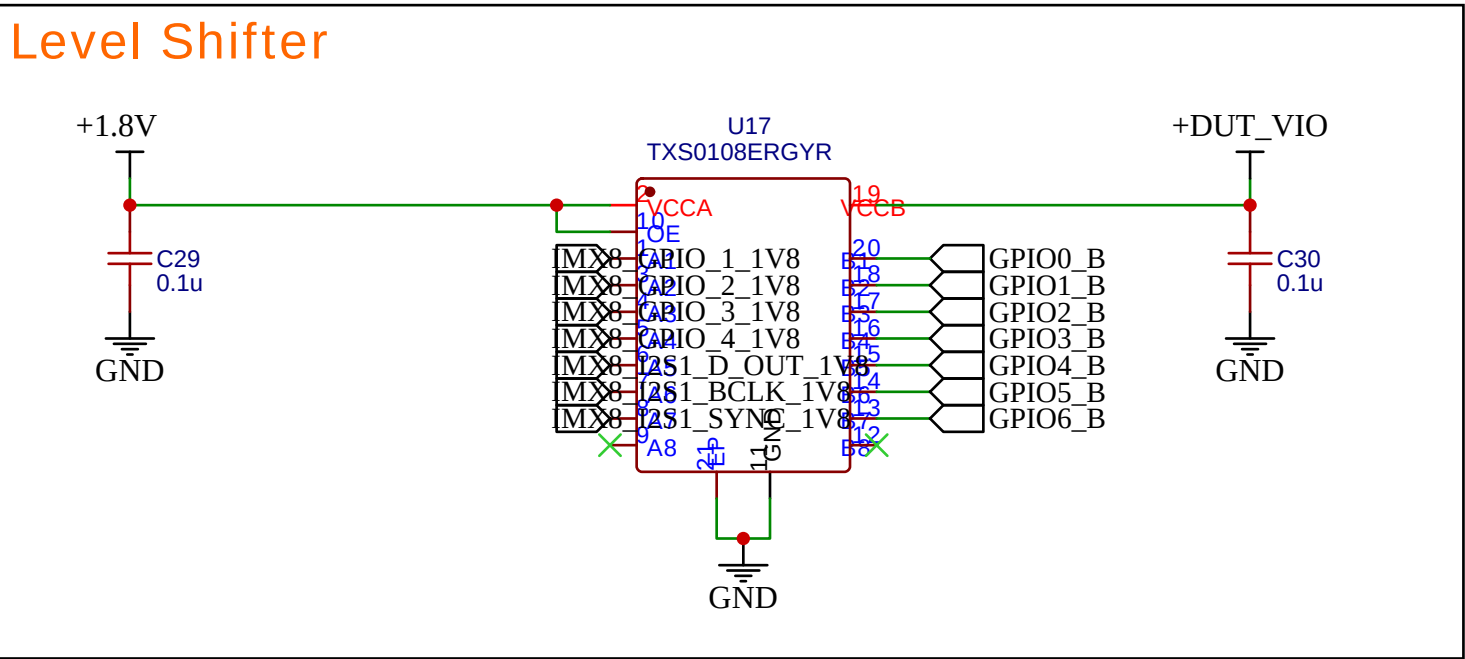
Internal Connections

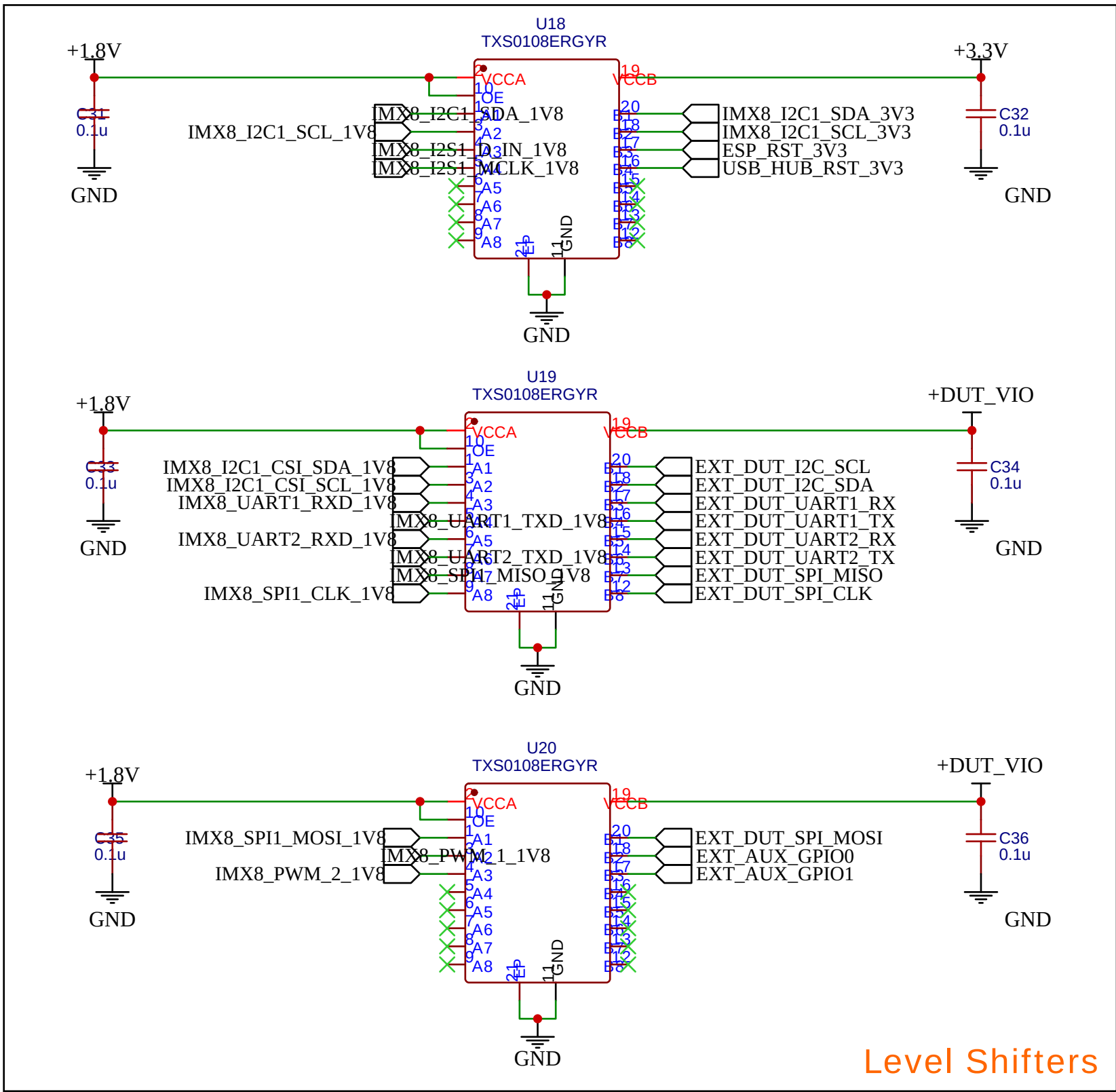
MAX_ADC_IN_V IMX8_ADC1_3V3

Test Points

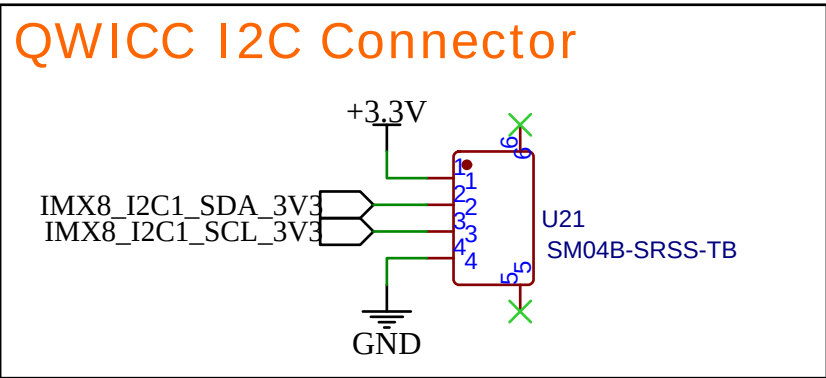
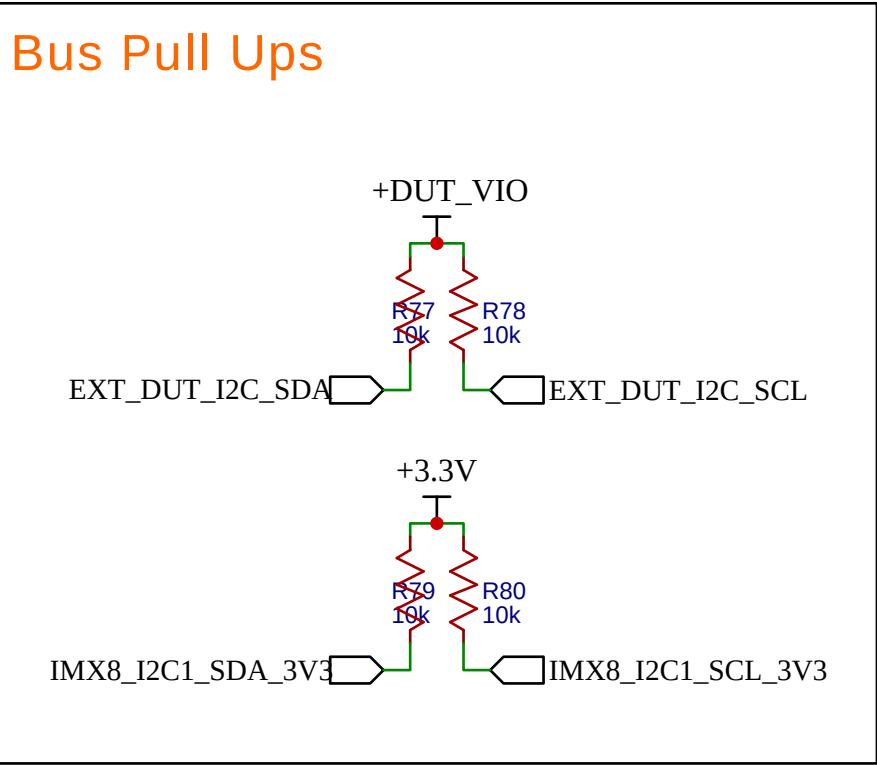
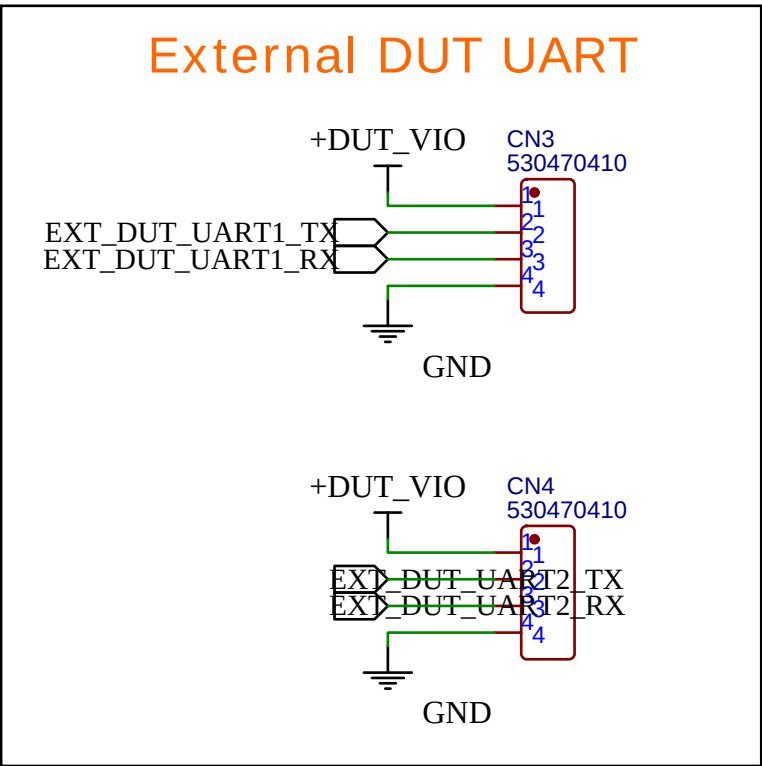
TP16 +16.4V
TP17 +14V_ADC

TITLE: Analog Signal Conditioning		REV: 1.0
COREKINET	Company: CoreKinect	Sheet: 7/14
	Date: 2/27/25	Drawn By: Mateo Segura

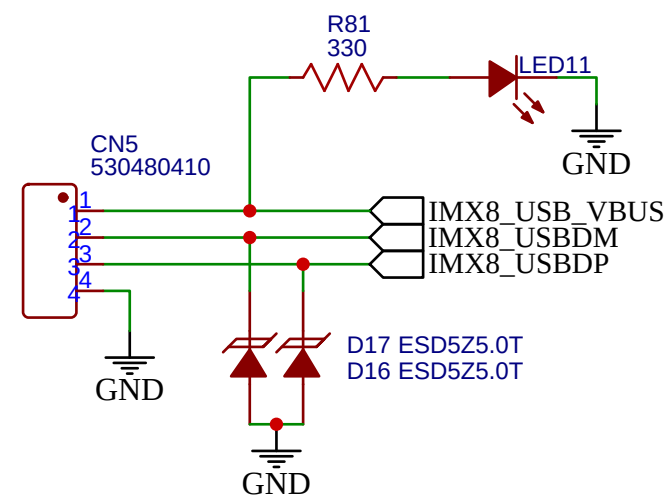




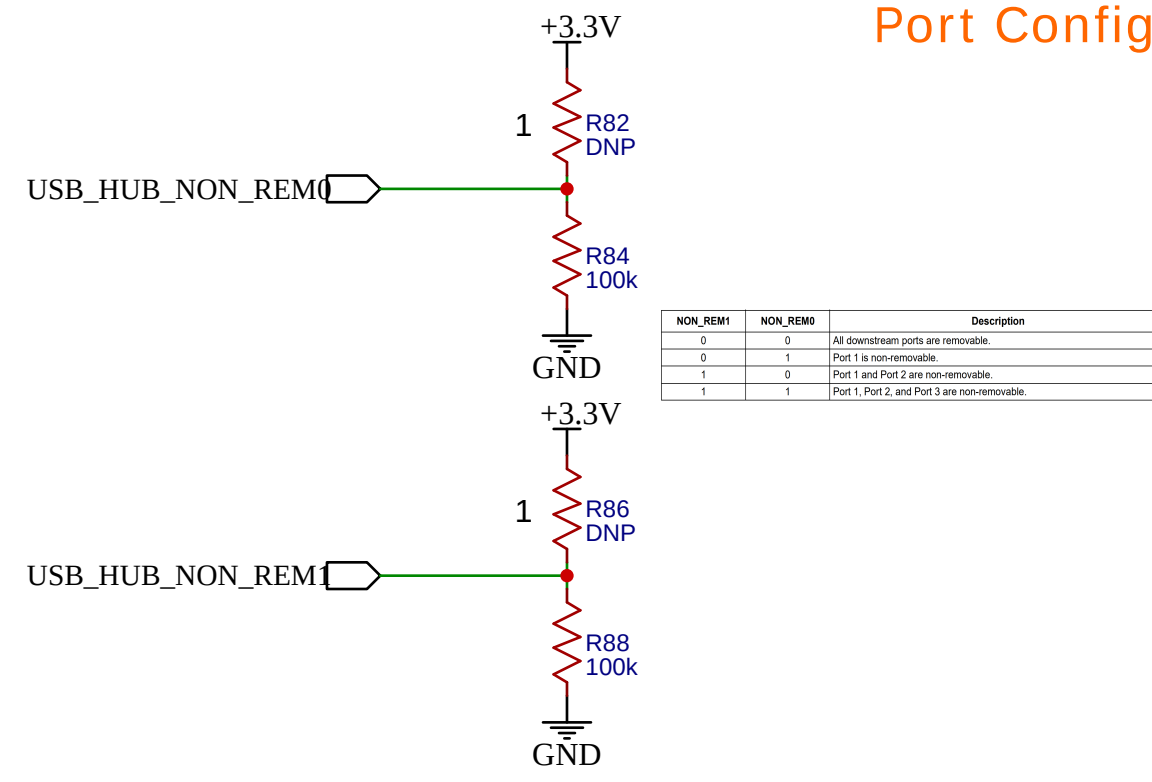
Level Shifters



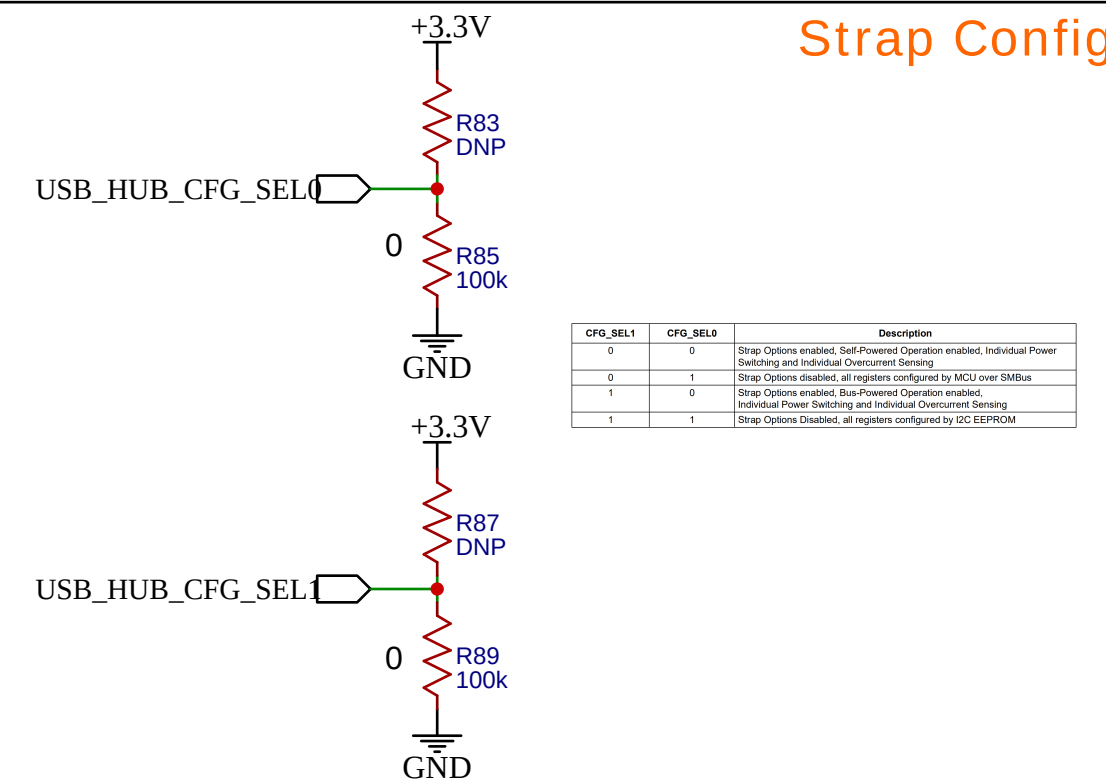
iMX8 Upstream Port



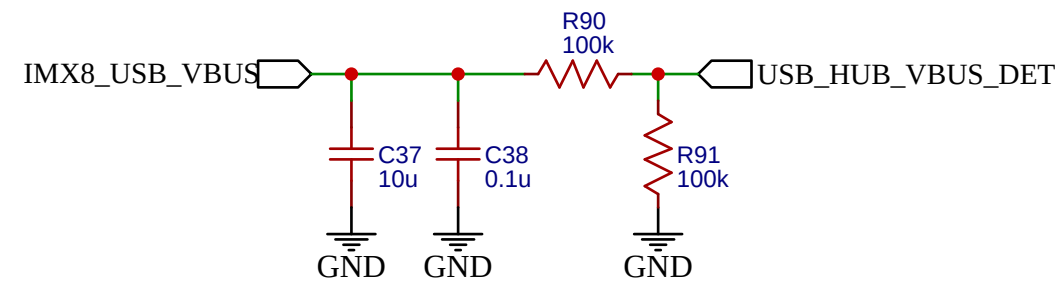
Port Config



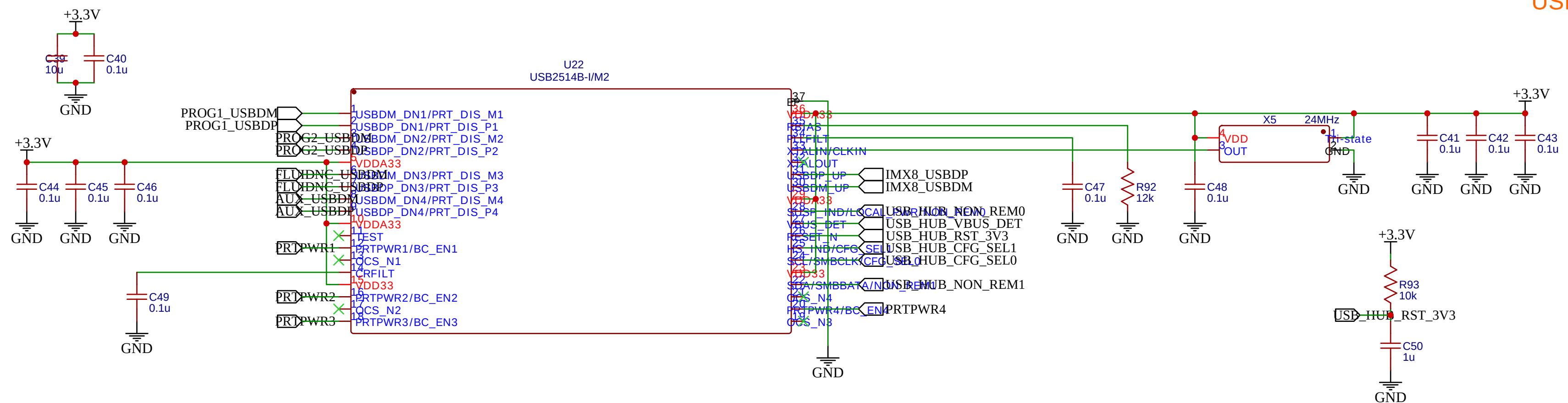
Strap Config



Bus Detection



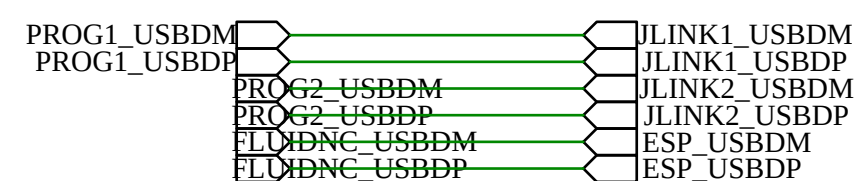
USB Hub



Notes

- USB DM and DP signals already have all necessary USB terminations and resistors in the IC
- VBUS_DET can be attached directly to 3.3V since it is not a detachable hub
- PRTPWR and OCS features are NOT used, as all devices and their power are known
- OCS pins have internal pull ups

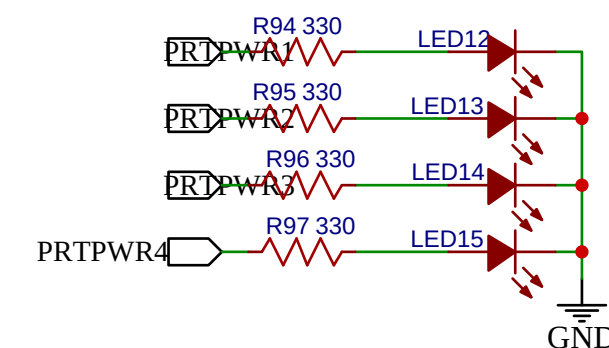
Internal Connections



External Connections



Port Power Debug LEDs

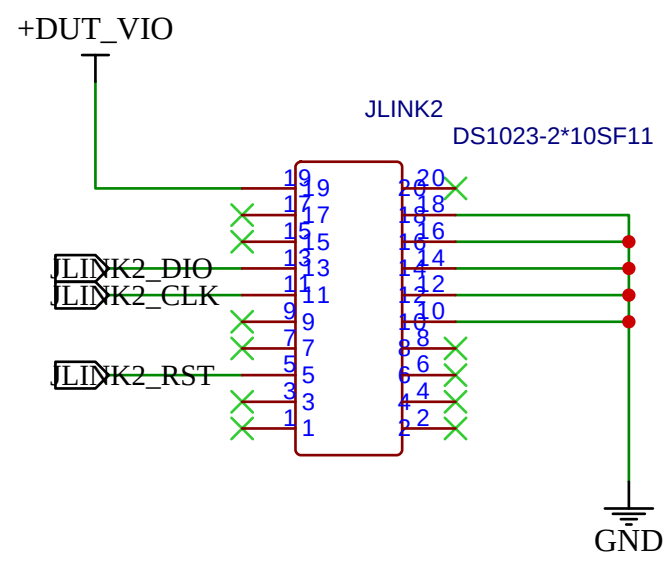
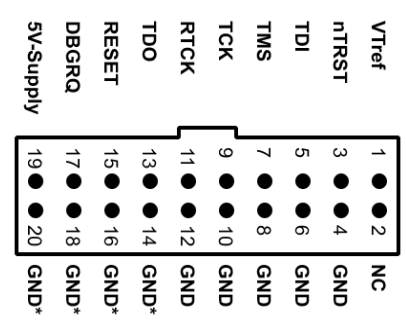
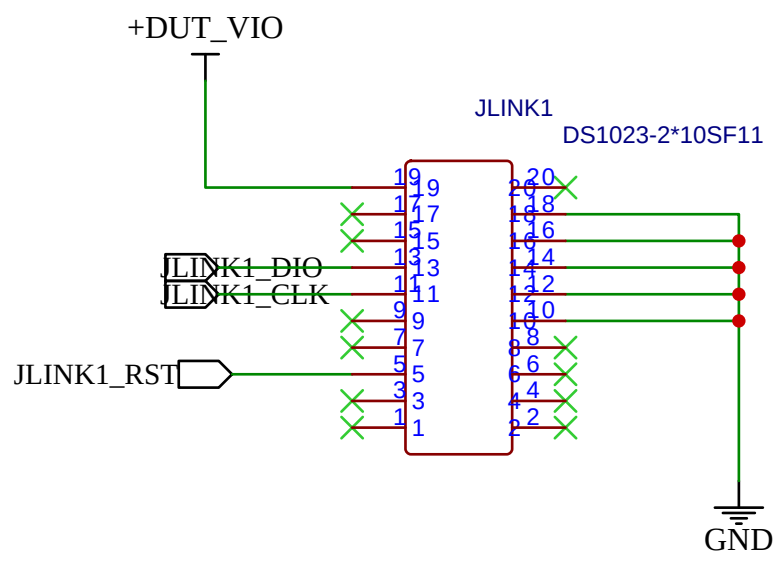


External USB

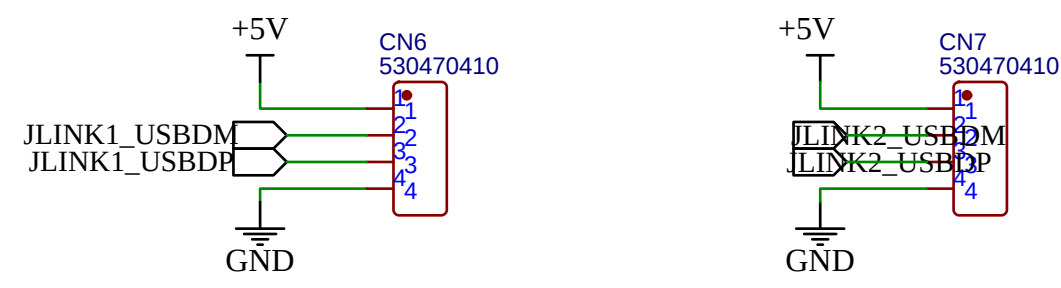


TITLE: USB		REV: 1.0
	Company: CoreKinect	Sheet:10/14
	Date: 2/27/25 Drawn By: Mateo Segura	

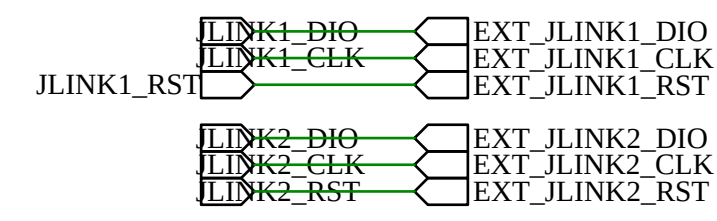
JLink Mini Connectors



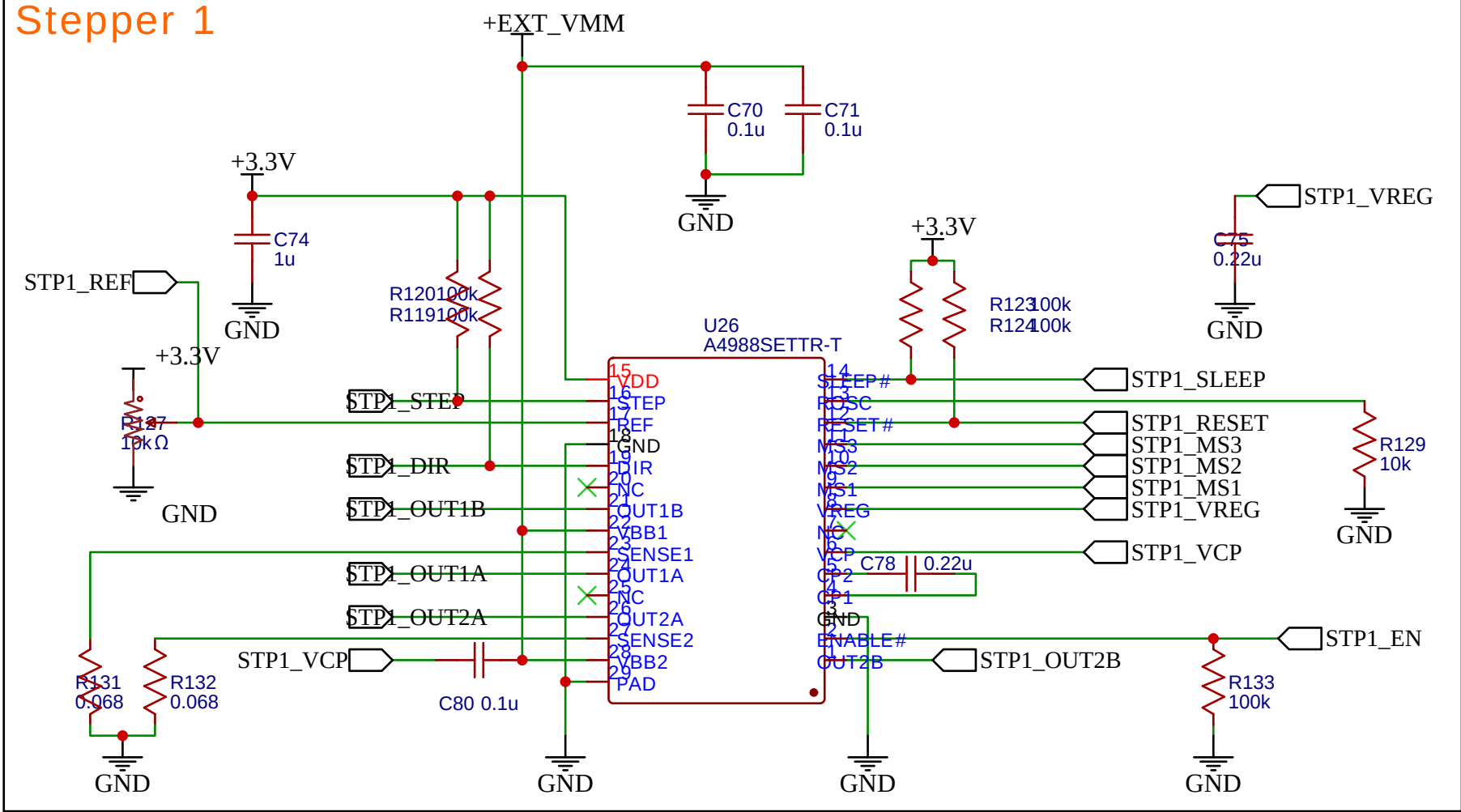
CoreKinect UART Style Connectors



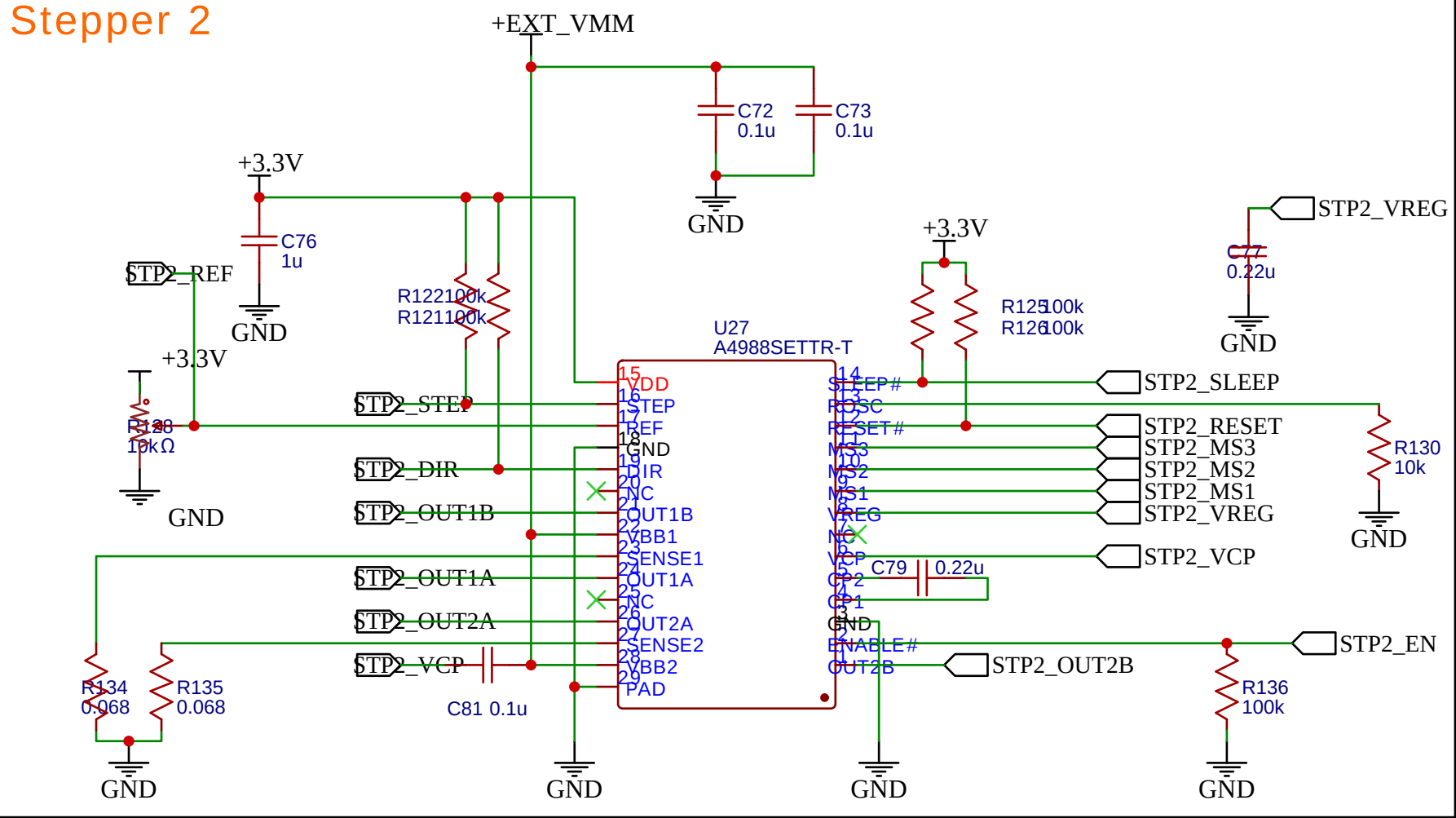
External Connections



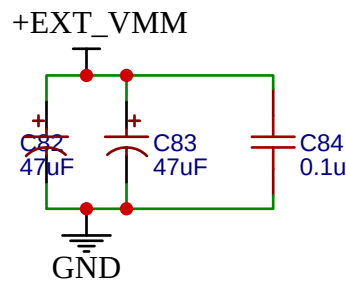
Stepper 1



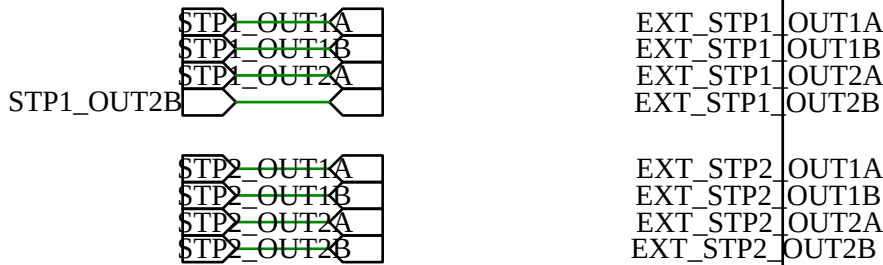
Stepper 2



Motor Power



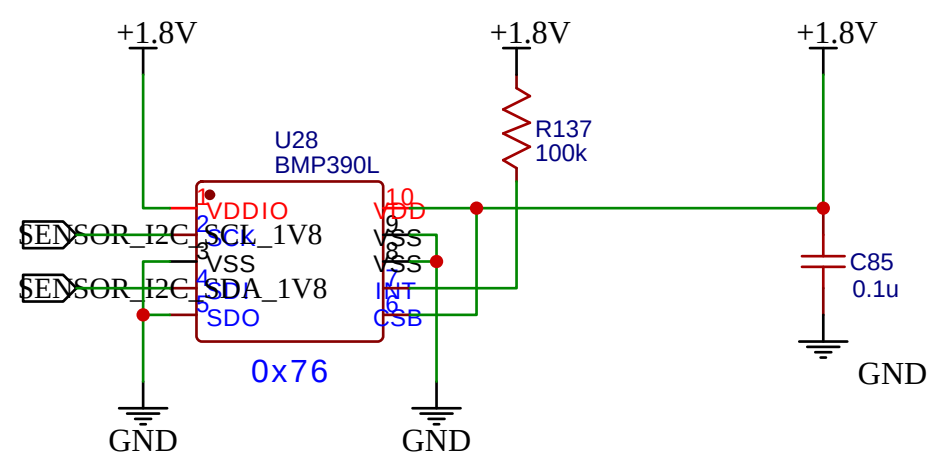
External Connections



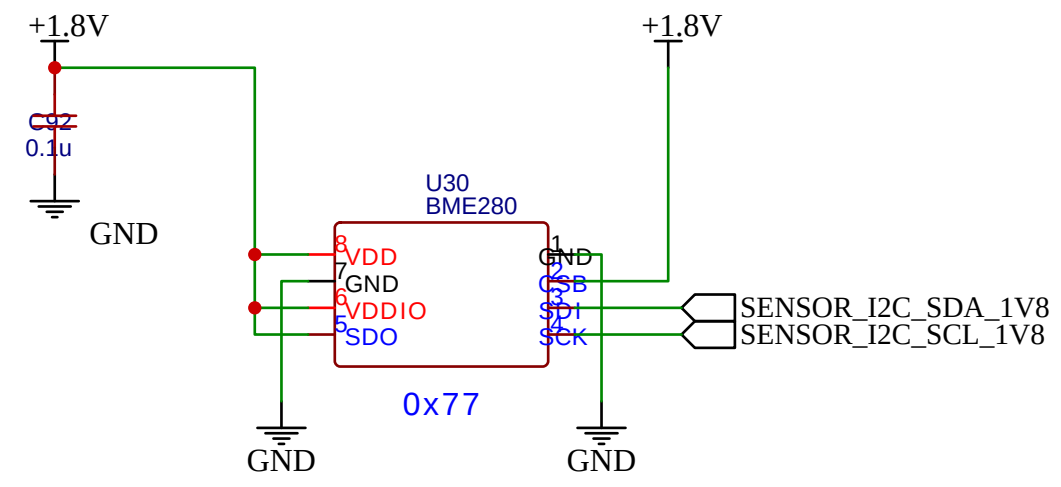
Test Points



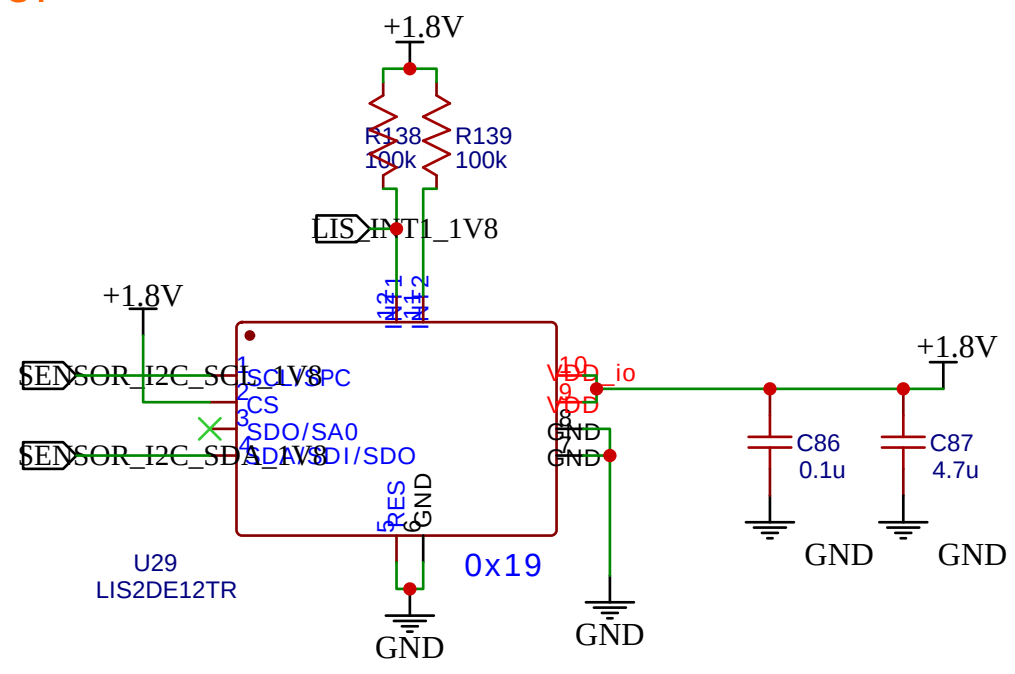
Pressure Sensor A



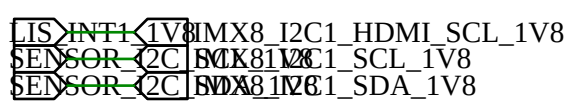
Pressure Sensor B



Accelerometer



Internal Connections



Connectors

- [x] Not all USB cables/connectors have the same polarity
- [x] CN5 should be switched to C588275

Power

- [x] MCP4017T-103E/LT was populated instead of MCP4017T-104E/LT
- [x] TPS Supply needs a 22uF cap on VOUT

ADC

- [x] Replace SPI ADC for 2x I2C ADC
- [x] ADC power LED is too bright
- [x] ADC Signal Conditioning Resistors are not sized correctly
- [x] INA on DUT Power can read voltage, so no need for ADC

Sensors

- [x] Add BME280

FluidNC

- [x] ESP32 pin 48 should NOT be connected to GND. Connect to a 10nF Cap instead

USB

- [x] Set all ports to be "removable" by setting R88 and R84

TITLE: ERRATA		REV: 1.0
COREKINECT	Company: CoreKinect	Sheet:14/14
	Date: 2/27/25 Drawn By:Mateo Segura	